(vi) Parametric form:

Parametric equations of a straight line is of the form

$$x = ar + x_1 \text{ and } y = br + y_1$$

where a and b are constants and r is the parameter.

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = r, (a \neq 0, \quad b \neq 0)\,.$$

Suppose we have the equation of the line passing through the point $Q\,(x_1, y_1)$ and making an angle θ with x-axis. Let $P\,(x, y)$ be a point on the line at a distance r from Q. Drop perpendiculars *QN* and *PM* respectively from Q and$P$ to the x-axis and perpendicular *QR* to *PM*.

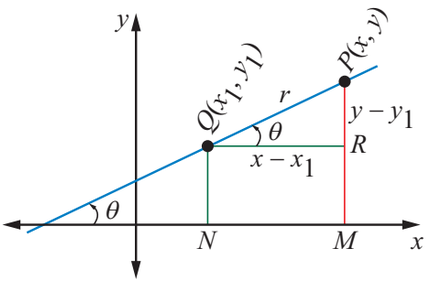


Figure 6.24

From the right ΔQRP

$$x - x_1 = QR = PQ\,\cos\theta \;=\; r\cos\theta$$

$$\text{Therefore } \frac{x - x_1}{\cos\theta} \;=\; r \qquad (6.10)$$

Similarly, $y - y_1 = RP = QP\,\sin\theta = r\sin\theta$

$$\Rightarrow \frac{y - y_1}{\sin\theta} = r \qquad (6.11)$$

From (6.10) and (6.11) we get

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r \qquad (6.12)$$

where the parameter r is the distance between (x_1, y_1) and any point (x, y) on the line. This is called the symmetric form or parametric form of the line.

The co-ordinates of any point on this line can be written as $(x_1 + r\cos\theta\,,\; y_1 + r\sin\theta)$. Clearly coordinates of the point depend on the value of r. This variable r is called parameter. Since r is a parameter the equations, $x = x_1 + r\cos\theta,\;\; y = y_1 + r\sin\theta$, is called the parametric equations of the line. The value of $'r'$ is positive for all points lying on the line one side of the given point and negative for all points lying on the line other side of the given point.

(vii) The general form of the equation of the straight line is

$$ax + by + c = 0, \quad \text{where } a,\; b \text{ and } c \text{ are all not zeros}$$